

Mentioned Is Not Binding: A Rules-plus-Small-Model Clause Ledger and a Pre-registered Audit of Regex Extraction on US Federal Solicitations

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Abstract

Compliance tools for US federal contracting commonly build a “clause ledger” by finding FAR and DFARS clause numbers in a solicitation with regular expressions. We ask how well that status quo answers the question a contracting team actually has: which of the clauses *bind* the contract? Without any annotator, we find that 13.8% of the entries such regexes produce over 14,540 extracted solicitation documents (58,294 of 423,328) are clauses whose every mention is a checklist item with an empty box, and 20–31% in checklist-heavy documents. We then build a rules layer (checkbox states, the unconditional paragraph (a) of FAR 52.212-5, inherited sub-item states, SF 1449 block 27, bare clause lists, exclusion vetoes) and a 0.46 MB mention-role classifier, and evaluate against the status quo in seven pre-registered rounds on documents not used in any earlier round, including a temporal hold-out of solicitations posted after the last acquisition day. Labels come from a coding agent and two OpenAI models, not human experts. The first two rounds, with a labelling protocol that had a defect, did not show an advantage and one pre-registered hypothesis failed; after the rules were extended on development documents, four consecutive fresh rounds show higher binding-set F1 than the status quo (+0.14, +0.10, +0.11, +0.08; 95% intervals above zero), specificity of 70–79% against 8–19%, and recall at or above the status quo’s in three of four rounds. We report negative results (weakly labelled retraining, a number-level stacker, the failed rounds, a review queue that never met its target), an annotator-agreement ceiling (pairwise annotator agreement 75–82%), and the main remaining error family. We release the model weights (pickle-free numpy archive with a numpy-only inference script), the per-mention table, all gold labels and round reports publicly.

1 Introduction

A US federal solicitation names its clauses in many ways: incorporated by reference, full text, checklists with marked boxes (FAR 52.212-5, agency checklists), bare lists, narrative citations, definitions inside other clauses, table-of-contents entries and construction specifications that cite clauses in running text. Regular expressions find clause numbers but not how each is used. The costs are asymmetric and product specific: a missed binding clause is a compliance risk; a false inclusion is review noise. We study the *ledger* task (which numbers bind a given document) and the *mention* task (the role of each occurrence).

Our contributions: (i) an annotator-free audit of the status-quo regex ledger on 14,540 documents; (ii) a small, fast system (rules plus a 0.46 MB logistic-regression role model, about 115 ms per document on CPU) with a single-document service; (iii) an evaluation that is explicit about what failed: pre-registered rounds, frozen artefacts, test documents disjoint across rounds including a temporal hold-out, and a record of every post-hoc change; (iv) negative results and an annotator-agreement ceiling that bound what can be claimed; (v) public model, data and reproducibility artefacts.

2 Related work

Contract NLP has benchmarks for provision classification (CUAD [1]; LEDGAR [2], also in LexGLUE [4]), document-level inference over contracts (ContractNLI [3]), and form layout understanding in scanned documents (FUNSD [5]). All classify *what a span says*; none asks whether a cited clause number *binds the contract*, which depends on checklists, incorporation lists and exclusion statements spread across the document. Long-context encoders such as ModernBERT [6] are the natural next modelling step (see §8); our shipped model is deliberately a small linear classifier so that the contribution of the symbolic rules layer can be measured cleanly. Pre-registration [7], model cards [8] and datasheets [9] structure our reporting; calibration follows [10] and intervals are bootstrap [11].

3 Data and task

Corpus. 4,755 solicitation notices from a federal-contracting API (posted 2024-10 to 2026-09) with their attachments (PDF, DOCX, DOC, TXT), 14,540 extracted documents, 6,606 with clause candidates; plus a temporal hold-out of 459 notices posted 2026-09-30 to 2026-10-07 (1,588 further documents). Candidate numbers come from regexes validated against a registry of 2,509 eCFR title-48 sections (version history, reserved and removed status, the 2025-10-28 FAR overhaul model deviation, and 47 DoD deviation memos); the model never generates a clause number. A layout-aware extractor keeps checkbox states (glyphs, form widgets, typed forms such as `__X__` (43) and `[X]`).

Ledger gold. For each sampled clause number a blind annotator answers: BINDS (incorporated by reference, full text included, a selected checklist item, or stated as applying), NOT, REFERENCED (a narrative requirement without incorporation), or UNDECIDED. The *binding set* is BINDS versus NOT/REFERENCED, undecided excluded. Annotators: the coding agent (hand labels from evidence sheets, before seeing any prediction), gpt-4o-mini and gpt-4.1-mini (instructions that state the checkbox convention and FAR 52.212-5(a)). The primary gold is the majority of the three. **No human expert annotated any item.**

Labelled universe. Numbers that have at least one mention no rule decides. Rule-decided numbers (checkbox items etc.) are objective and reported separately.

4 Method

Status quo (B0). A port of the production regexes, shown equal to the production PHP on 403 cases. **B1** is a section-aware rules baseline.

Rules layer (v1.4). (1) Checkbox state of an item line (glyph, widget, typed form). (2) The unconditional paragraph (a) of FAR 52.212-5 is binding. (3) A sub-item whose marker was lost inherits its parent line’s state. (4) SF 1449 blocks 27a/27b by their leading glyph. (5) Wordings that are not incorporation by themselves: “applies only if the clause at... is included”, “has the meaning provided in the clause at...”, a paragraph reference, “or FAR x, as applicable”, “refer to the clause at...”; and “FAR Clause x...is applicable for this solicitation” as binding. (6) Bare lists of at least four clause numbers with titles, outside a table of contents. (7) Explicit deletion/exclusion wordings (“CLAUSE x IS DELETED”, “x does not apply”) veto the number through the ledger aggregation.

Role model. One-vs-rest logistic regression over about 60 structural/wording features plus character TF-IDF of the context, temperature scaling (development ECE 0.06), trained on 2,663 mentions from 79 documents (mostly rule-labelled checkbox items). A number’s probability is the maximum over its mentions; it binds at $q \geq 0.5$. The weights have not changed since the first test; the public archive is a pickle-free numpy export verified to match the training-time model to 10^{-9} .

Service. `f1 ledger FILE` returns the clause ledger for one document (tier, probability, evidence lines, registry status, optional version currency) in about 2.5 s for a 42-page PDF on CPU.

5 Evaluation protocol

Each round fixes in writing, before any label, the frame (sampling script and seed, sha256 of the document list), the primary configuration and the numeric hypotheses; artefacts are frozen by hash; documents and solicitation numbers are disjoint from training, development and all earlier rounds. Metrics: binding-set F1 over the labelled universe, micro over documents, cluster bootstrap over documents (10,000 resamples), paired bootstrap for differences. Rounds 2 to 7 (and the corrections described below) are *development data* for any change made after they were read; only a later round is a test of that change. Round 7 samples only the temporal hold-out.

6 Results

Annotator-free audit. Of 423,328 status-quo entries, 58,294 (13.8%) are numbers all of whose mentions are checklist items with an empty box, in 2,390 of 3,029 documents with a decided checklist (median 30 per affected document). In checklist-heavy documents the share is 20.6–31.0% across rounds (round 7: 22.9% on the temporal hold-out). The rule that reads these boxes agreed with a reader on 300 of 300 random items over five rounds (60 per round).

What failed. Round 1 (32 documents, as pre-registered) showed non-inferiority only (+0.020 [−0.005, +0.046]); the hypotheses on specificity and on abstention failed, and after the test we found a label-letter collision in the judge prompt (the letter R occurs in documents) and corrected it post hoc. Round 2 (36 documents, checklist-heavy) failed its non-inferiority hypothesis (−0.103 [−0.185, −0.036]) because the two OpenAI judges accepted unchecked boxes as binding (Cohen κ 0.05 and 0.21 against the agent); with instructions stating the checkbox convention (post hoc) the difference was +0.047 [−0.000, +0.096].

Fresh tests. After reading the errors of rounds 2 and 3 the rules were extended (v1.2), after rounds 4 and 5 again (v1.3: bare lists, alternative references), and after round 6 again (v1.4: exclusion vetoes, spec-reference wording); each version was then tested on documents never read before, round 7 on the temporal hold-out. In rounds 4, 5 and 7 every annotator’s point difference

Test	Docs	Nums	F1 B0	F1 model	Δ F1 [95% CI]	Spec B0	Spec model	Rec B0	Rec model
Round 3 (rules v1.1)	30	553	0.778	0.890	+0.111 [+0.049, +0.181]	0.16	0.84	0.933	0.874
Round 4 (rules v1.2)	26	519	0.780	0.921	+0.142 [+0.085, +0.210]	0.10	0.76	0.877	0.940
Round 5 (rules v1.2)	24	395	0.829	0.922	+0.093 [+0.034, +0.155]	0.19	0.79	0.899	0.916
Round 6 (rules v1.3)	24	465	0.807	0.914	+0.107 [+0.057, +0.161]	0.08	0.72	0.939	0.939
Round 7 (rules v1.4, temporal)	27	485	0.806	0.890	+0.084 [+0.028, +0.143]	0.14	0.70	0.924	0.903

Table 1: Majority-gold results of the five tests of the rules layer on fresh documents (model: max over mentions, $q \geq 0.5$; B0: status-quo regexes). Δ F1 with a 95% cluster-bootstrap interval. Round 3 tested v1.1 and had a recall cost; rounds 4 to 7 tested v1.2–v1.4, round 7 on the temporal hold-out.

is positive; in round 6 every interval is above zero; in round 3 the judge intervals include zero. H2 (specificity ≥ 0.70 with recall ≥ 0.90) holds in rounds 4–7, with thin margins in round 7 (0.703, 0.903). Gains concentrate in checklist-heavy documents (large stratum, round 4: +0.33 F1) and are small in short documents without checklists (+0.03).

Ceiling. Annotators agree with each other on 75–82% of binding-set items (κ 0.47–0.56, Table 2). The model agrees with the agent on 95% (κ 0.88; the agent also wrote the rules), with gpt-4o-mini on 80% and with gpt-4.1-mini on 73%. On items where two annotators agree, the left-out annotator matches their consensus in 81–89% of cases, versus 87–96% for the model depending on whose consensus is used (the model’s number is inflated where the agent trained it). A score of 0.98 on all criteria is therefore not attainable against these labels without fitting to annotator-specific conventions.

Pair	Items	Agreement	Cohen κ
agent vs gpt-4o-mini	1958	0.821	0.56
agent vs gpt-4.1-mini	1807	0.760	0.49
gpt-4o-mini vs gpt-4.1-mini	1799	0.752	0.47
model vs agent	2018	0.950	0.88
model vs gpt-4o-mini	1981	0.805	0.51
model vs gpt-4.1-mini	1821	0.730	0.43

Table 2: Pairwise agreement on binding-set labels pooled over rounds 3 to 6.

Negative results. (a) Retraining on 2,404 weak labels where two OpenAI voters agreed scored below the rule-only version on development data in all three variants. (b) A number-level stacker (logistic regression over mention probabilities, trained on the agent’s labels of rounds 2 to 4) had cross-validated AUC 0.982 against 0.949 for max-q but gave +0.004 [−0.003, +0.014] F1 on a fresh round, so max-q remains the default. (c) A calibrated review queue (probability in (0.1, 0.9)) caught 58%, 48%, 64% and 53% of the errors in four rounds (targets were 60%).

Remaining errors. Construction specifications that cite clauses in running text (clean errors of rounds 4–7 are dominated by a few specification documents), clause lists without markers,

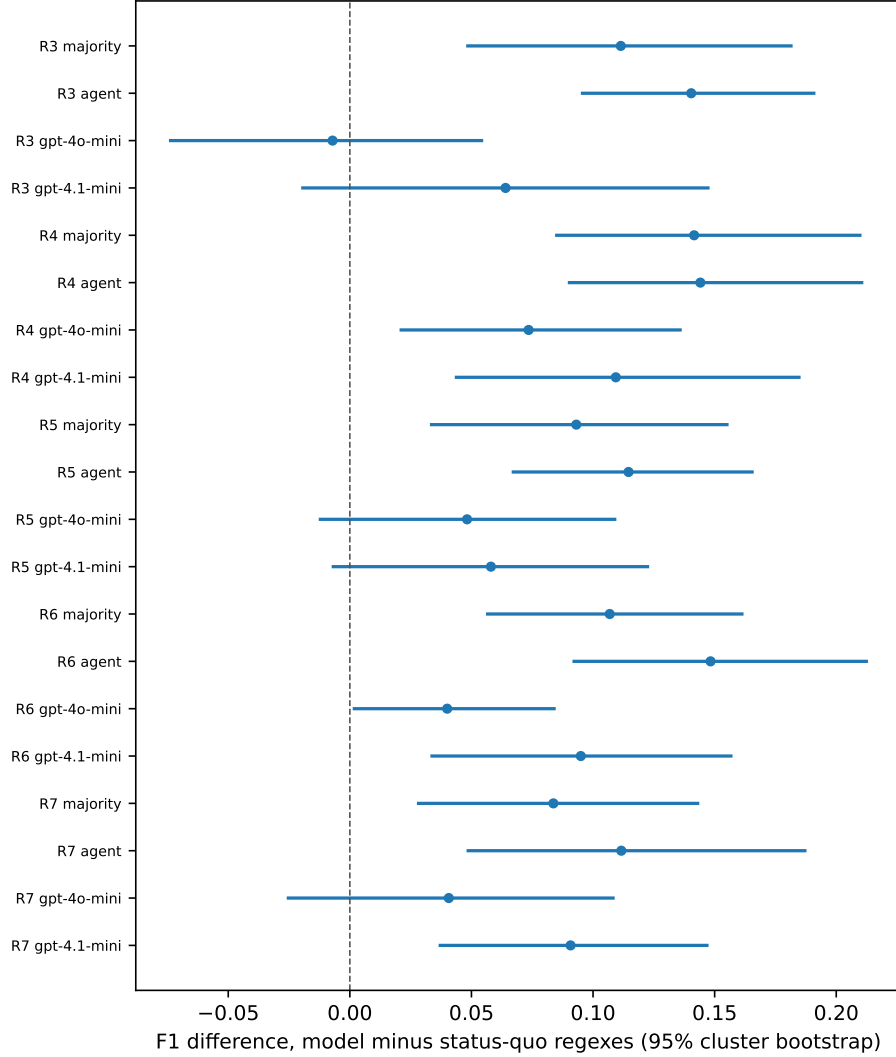


Figure 1: F1 difference (model minus status quo) per test and annotator, with 95% cluster-bootstrap intervals.

documents whose checkbox markers were lost in extraction (labelled UNDECIDED), and exclusion wordings placed mid-line after the clause title (found while labelling round 7; queued for v1.5).

7 Limitations

(1) No human expert labels; the judge instructions derive from the agent’s conventions, so the annotators are not independent readers of the format. (2) The agent that labelled also wrote the rules; rounds 2 to 7 are not independent of rule design except through freshness of documents. (3) 24–36 documents per round (27 in round 7, drawn from a one-week posting window) with recurring federal templates; the rounds are not independent draws of the template population. (4) Numbers only; alternates, dates and ranges are not in the gold. (5) A federal-contracting API sample from 2024-10 to 2026-10; one scanned-document share ($\approx 3\%$) is not processed. (6) Not legal advice; the model must not remove clauses silently.

8 Roadmap

The decision log queues: an encoder fine-tune (ModernBERT-base span classifier over layout-marked windows, the Phase-8 design the linear model stands in for), an “applicable” mode for narrative requirements (the REFERENCED tier, where B0 still leads), review-queue calibration, and more document templates (SF 30 amendments, lease clauses). Reinforcement learning is not on this list: the shipped system is a classifier with a fixed label schema, and its errors are concentrated in document-structure handling that rules and supervised span classification address directly; RL would first need a reward signal that does not exist (no expert preference data) and would inherit the same annotator ceiling.

9 Release, ethics and reproducibility

Documents are public SAM.gov attachments. Extraction redacts personal-data patterns (contact emails and phone numbers, hardened against glued tokens; verified zero remaining in the release). The release contains derived tables and short snippets, not documents. Every number is generated from `results/*.json` by scripts in the repository; decisions are logged (D-001 to D-045). Public artefacts: model weights as a pickle-free numpy archive with a numpy-only inference script (<https://huggingface.co/raihan-js/fedproc-ledger-v1>, Apache-2.0), dataset and round reports (<https://huggingface.co/datasets/raihan-js/fedproc-ledger-bench>). No Qwen or other non-US model was used. The authors are affiliated with a company that sells proposal software; this work uses only public solicitations and the integration contract keeps customer documents in-boundary.

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